Politecnico di Torino

Lab report for the course Qubit Electronics

Master degree in Quantum Engineering

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CHAPTER 1

Introduction

The objective of this laboratory activity is to characterize the functionality of a device through a black-box approach. In fact, using a Vector Network Analyzer (VNA), we measured the scattering parameters of a device whose actual application was unknown. Then, by interpreting the results, we inferred its functionality.

1.1 Scattering matrix

The coefficients of the scattering matrix allow us to describe the behavior of a device in terms of transmission and reflection of an input signal. Considering a two-port device, as shown in Fig. 1.1, and without knowing how the device is internally implemented, we focus only on what happens at its ports.

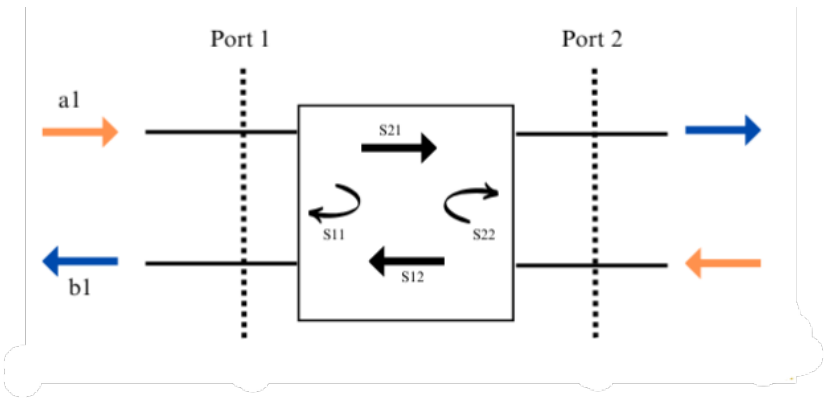

Figure 1.1: Representation of the two-port device.

Let us define a_i as the incident wave and b_i as the reflected wave, with $i = 1, 2$ representing port 1 and port 2, respectively. We can write the outgoing waves at the ports as:

$$b_1 = S_{11}a_1 + S_{12}a_2$$
$$b_2 = S_{21}a_1 + S_{22}a_2$$

These relations can be expressed in matrix form, where **a** and **b** are vectors and the S_{ij} terms are the coefficients of the so-called scattering matrix.

$$\begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$$

In particular, the index i represents the output port, while j represents the input port.
These values are defined as

$$S_{ij} = \left. \frac{b_i}{a_j} \right|_{a_k=0,\ k\neq j}$$

by setting to zero the contributions of the incident waves from all the other ports except the one being considered.

The diagonal terms, S_{11} and S_{22}, represent the reflection coefficients of the respective ports, while the off-diagonal terms, S_{21} and S_{12}, describe the transmission between the two ports.

1.2 Measurement setup and calibration

To set up the measurement environment, we used a Keysight VNA together with its dedicated software. First of all, we configured the plots of the S-parameters that would be observed during the measurements. In particular, we used two windows: the first displaying the S_{11} and S_{22} parameters, and the second displaying S_{12} and S_{21}. Observing all four S-parameters allowed us to evaluate both the reflection and transmission behavior of the unknown device.
The frequency range for the entire laboratory session was set from 10 MHz to 6.5 GHz, with a resolution of 1001 points uniformly distributed between the two edge values.
To minimize the effect of random noise during the measurements, we enabled a software setting that allows multiple acquisitions to be performed and then averaged in the displayed result. In our case, we used an average over 8 measurements.

Before starting the actual measurements, we performed the VNA calibration. The calibration procedure is important for removing systematic errors that may occur for several reasons, such as attenuation or phase shifts introduced by the cables connecting the DUT to the VNA, as well as distortions caused by the internal couplers of the instrument. To calibrate the VNA, we used an **Electronic Calibration kit** (E-cal kit), a device that is connected to the VNA and is able to emulate standards such as an open circuit, short circuit, precision load, and a through connection. After connecting the E-Cal kit to the VNA, we started the calibration procedure through the software. This procedure consists of measuring the known standards emulated by the E-Cal kit and then using them as a reference to correct the measurements, thereby eliminating systematic errors.

CHAPTER 2

Passband Filter

2.1 Setup

The device, showed in figure 2.1, has two ports which have been connected to the VNA ports. By looking at the device's response, we focused on the range 1.5GHz-3GHz, which is the one worth to be analyzed. The results have been extracted as Touchstone files and then they have been elaborated using python, in particular using the `scikit-rf` module.

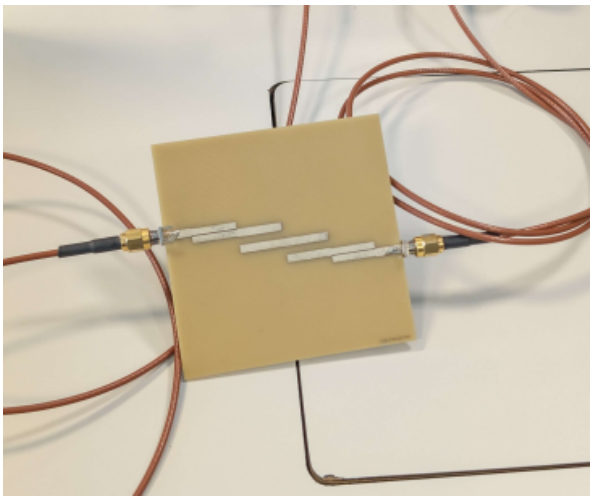

Figure 2.1: Image of the device connected to the VNA

2.2 Analysis of the measurements

By looking at figure 2.2, which shows the magnitude of the signal of the respective channel, is it clear that the device has a reciprocal response between the two ports, since the plots of $S11$,$S22$ are identical as well as $S12$,$S21$ plots. The validity of this assumption is confirmed by the extraced values showed in table 2.1. This is a typical behavior of a passive component and so, for the rest of the analysis, we will focus only on the $S22$ and $S12$ parameters (second row of the figure).

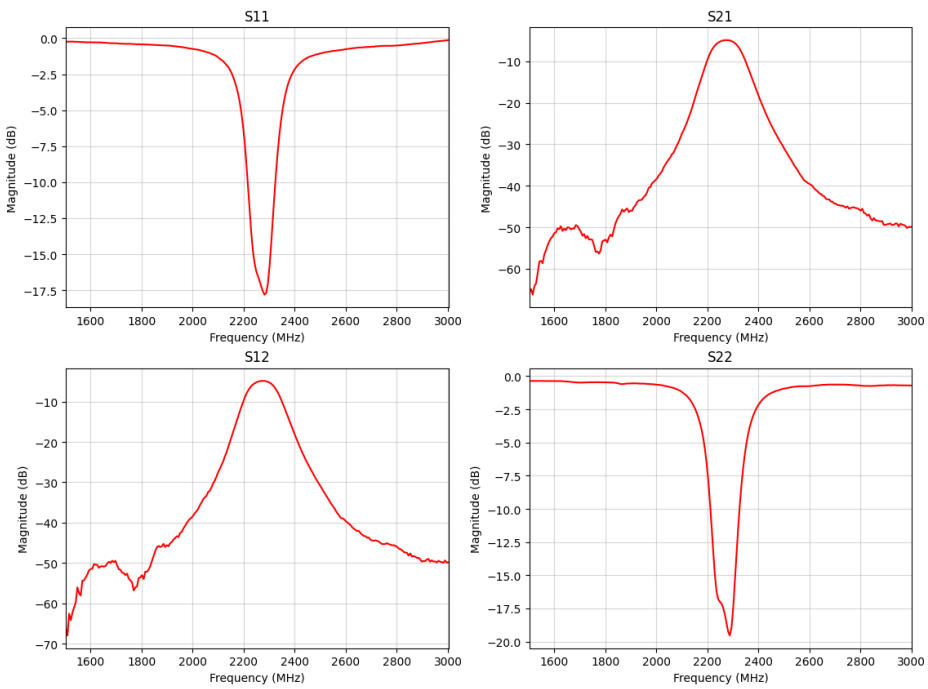

Figure 2.2: Plots of the parameters of the Scattering matrix

P	f_{res} **[MHz]**	**Min. [dB]**	**MBW [MHz]**
S_{22}	2287	-19.53	103.84
S_{11}	2281	-17.81	103.84

P	f_c **[MHz]**	**Max. [dB]**	**BW [MHz]**
S_{12}	2275	-4.88	123.31
S_{21}	2275	-4.88	123.31

Table 2.1: Values of the S parameters, where f_{res} is the resonance frequency, MBW is the matching bandwidth calculated at < -10 dB, f_c is the central frequency, and BW is the bandwidth calculated at -3 dB. The reciprocity of the device is evident from the symmetry between S_{12} and S_{21}.

By examinating the $S12$ parameter, we understand that the device is allowing the signal to pass from input port (port 2) to output port (port 1) in a frequency range around 2275MHz, with a bandwidth of 123.31MHz. This tells us that the device has a quite wide bandwidth, making it suitable for the transmission of a not so limited number of frequencies. At peak frequency, the signal has an attenuation of -4.88dB, meaning that when the transmitted signal is at its peak, almost 60% of it is transmitted. This is due to the fact that the device is not an ideal one so it is affected by internal losses.
Regarding the $S22$ parameter, we observe that the signal is almost completely reflected at frequencies below 2200MHz and above 2300MHz, while the reflection is minimum between these values at 2275MHz where the magnitude of the reflected signal corresponds to -19.53dB. This behaviour is consistent with the previous analysis of the $S12$ parameter, since the signal is transmitted to port 1 from port 2 at the same frequency range at wich, on port 2, the signal is not reflected.

2.3 Summary

In conclusion, we can state that the device under test acts as a two-ports reciprocal passband filter. This conclusion is supported by the transmission bandwidth (S_{12} or S_{21}), which is sufficiently wide to exclude the possibility of a high-Q resonator. Given the operating frequency range around 2300MHz,

the device could be used in wireless communication systems, such as S-band applications or wireless data links for filtering out unwanted frequencies.

CHAPTER 3

Antenna

3.1 Setup

The DUT is shown in figure 3.1. It has a single port which has been connected to the VNA port (port 1 in this case). Since in this case we have only one port, the study will be focused on the $S11$ parameter, which is the only one available, and so on the behavior of the reflected signal.

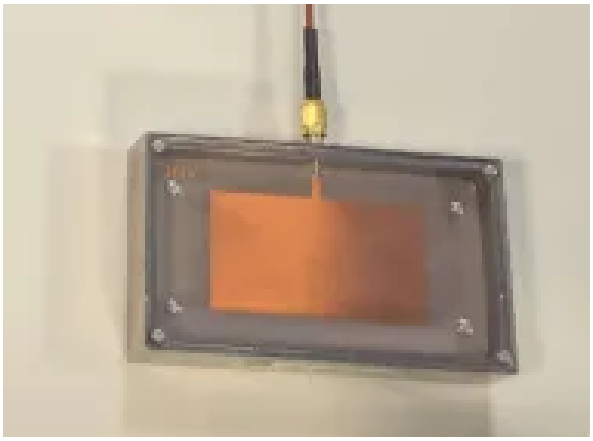

Figure 3.1: Image of the device connected to the VNA

3.2 Analysis of the measurements

From the plot illustrated in figure 3.2 we can observe two dips in the magnitude of the reflected signal at a frequencies of $f_{res} = 3001.89$ MHz and $f_{res} = 6058.68$ MHz. At this frequencies, the signal has a return loss of -11.46dB and -16.34dB respectively, with a matching bandwidth of $BW = 6.49$ MHz and $BW = 84.37$ MHz (calculated at < -10 dB threshold). When we reach frequencies around 3500 MHz, the measurements show an unstable behavior where the magnitude exceeds 0 dB. This represents an "unphysical gain" and is likely a calibration artifact or the result of a poor connection between the device and the VNA port during the setup phase.
Focussing now on the operating frequenices of the device, which comprehends signals around 300 and 6000 MHz, the return losses at these resonance frequencies indicate that 92.8% of the incident signal in the first case, and 97.67% in the second case, is not reflected, but instead is transmitted by the device. Since we have only one port, we can assume that the signal is transmitted by irradiation in

the space, meaning that the device is acting like an antenna at the resonance frequency of 3001.89 and 6058.68 MHz.
By considering the matching bandwiths of 6.49 and 84.37 MHz, which leads to a very high quality factors of $Q = 462.5$ and $Q = 71.4$, we can state that the device has a high selectivity on the transmitted frequencies but also a very low sensitivity to the variations of the frequency of the transmitted signal. This makes the DUT also not too reliable, since a small variation also due to external noise can lead to a significant decrease of the transmitted signal.

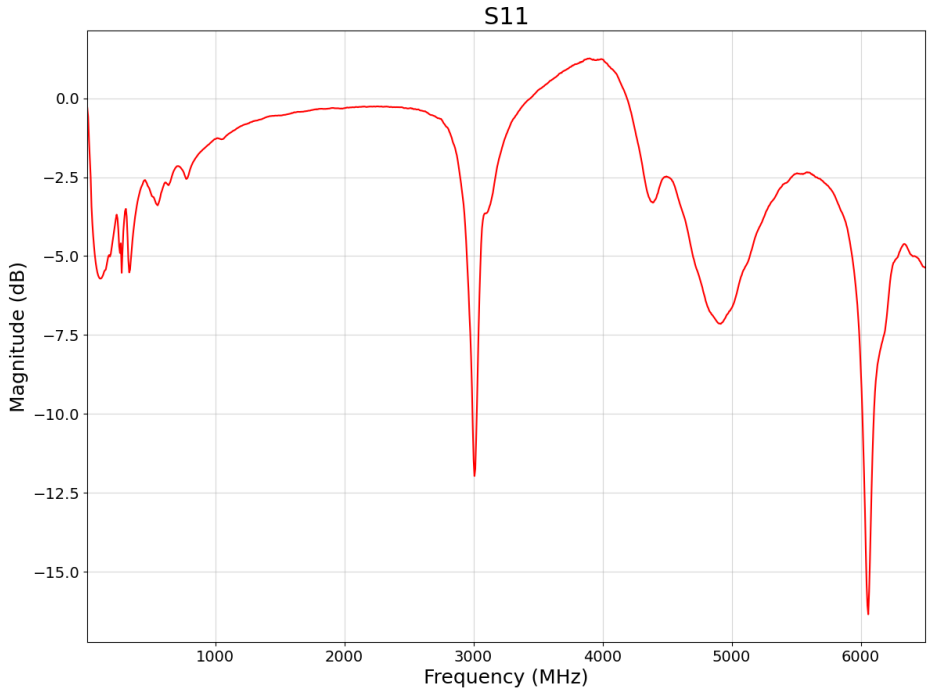

Figure 3.2: Plot of the $S11$ parameter of S matrix

3.3 Summary

In conclusion, the DUT in this case acts as a single-port dual-band antenna which transmits signals at a frequency of around 3000 and 6000 MHz with a very low matching bandwidths and so a high selectivity of the transmitted frequencies. An interesting observation is that if a hand is placed close to the surface of the device, we could see a shift of the dip towards lower frequencies. This phenomenon is due to the capacitive coupling between the hand and the device, which affects the resonant frequency of the antenna.

CHAPTER 4

Circulator

4.1 Setup

The DUT is a three-port circulator, a non-reciprocal RF device that routes signals directionally between its ports. Ideally, the circulator enforces a cyclic transmission of power:

$$1 \to 2, \quad 2 \to 3, \quad 3 \to 1$$

This means that a signal entering one port is transmitted only to the next port in the sequence, while the other port is ideally isolated.

Real circulators exhibit non-idealities such as insertion loss in the transmission path and finite isolation between non-connected ports.

In order to characterize the device, two configurations were measured using a VNA:

- **Configuration 1**: ports 1 and 2 connected to the VNA, with port 3 terminated by a matched $50\,\Omega$ load
- **Configuration 2**: ports 1 and 3 connected to the VNA, with port 2 terminated by a matched $50\,\Omega$ load

The matched termination is necessary to absorb the incoming power and avoid additional reflections from the unused port, which would otherwise modify the measured response.

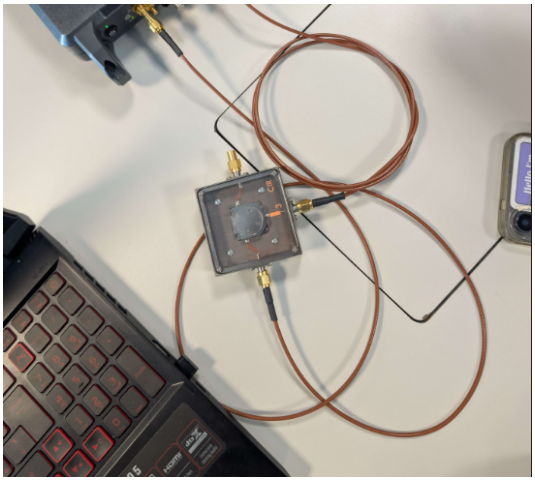

Figure 4.1: Experimental setup used for the circulator characterization.

4.2 Analysis of the Measurements

4.2.1 Configuration 1

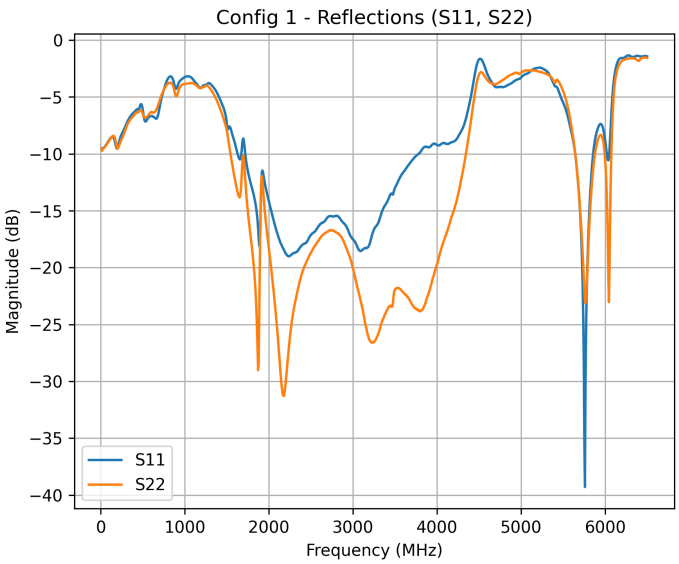

Figure 4.2: Reflection coefficients S_{11} and S_{22} for configuration 1.

The reflection coefficients S_{11} and S_{22} are generally below $-10\,\mathrm{dB}$ over a wide portion of the frequency range, indicating a reasonably good impedance matching at both measured ports. More pronounced minima are visible around the 2–3 GHz region, corresponding to frequencies where the device is better matched and therefore more efficient in transmitting power.

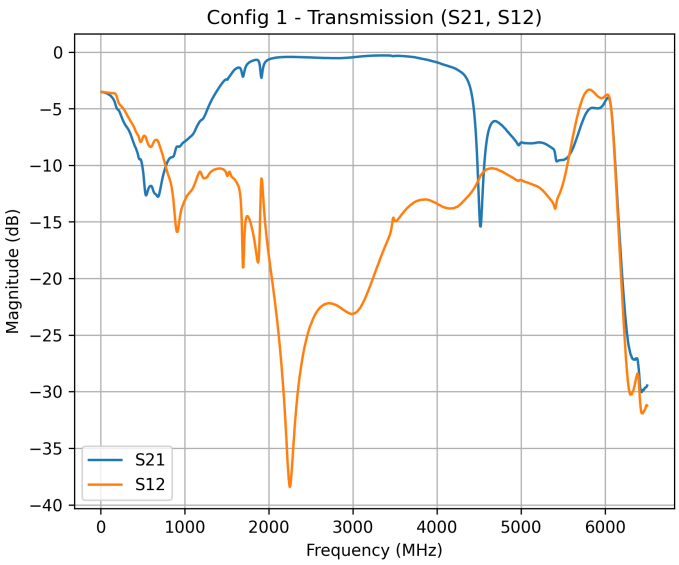

Figure 4.3: Transmission coefficients S_{21} and S_{12} for configuration 1.

The transmission response clearly shows the non-reciprocal behavior of the device. In this config-

uration, S_{21} remains close to 0 dB over a broad frequency interval, meaning that power injected into port 1 is efficiently transferred to port 2. On the contrary, S_{12} is much smaller and reaches strongly attenuated values, showing that the reverse propagation is suppressed. This is the expected behavior for the allowed transmission path $1 \to 2$.

4.2.2 Configuration 2

Figure 4.4: Reflection coefficients S_{11} and S_{22} for configuration 2.

Also in the second configuration the reflection coefficients indicate a generally acceptable matching, with some frequency regions showing deeper minima and therefore better impedance adaptation.

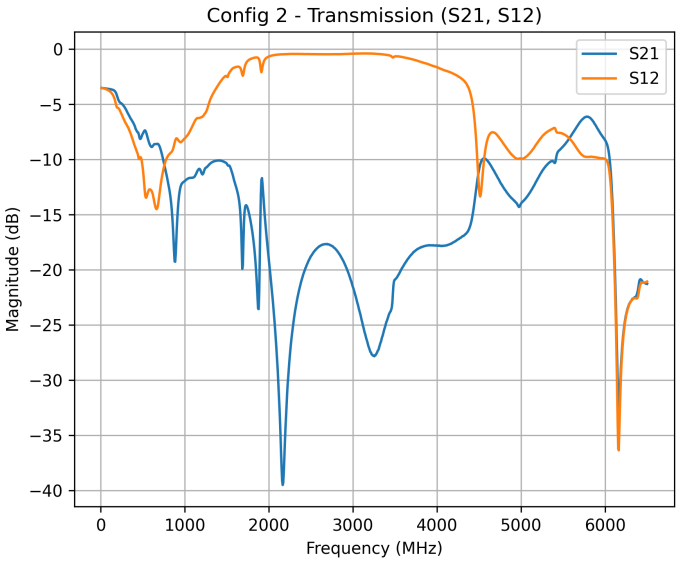

Figure 4.5: Transmission coefficients S_{21} and S_{12} for configuration 2.

In this case the connection of the ports is changed, and the measurements again highlight the directional nature of the circulator. One transmission coefficient is significantly larger than the opposite one over the useful frequency range, while the reverse direction remains attenuated. This confirms that the device does not behave reciprocally and that the transmitted power depends on the ordering of the ports. terminating the third port with a matched $50\,\Omega$ load, it is possible to directly observe the cyclic routing of signals that characterizes the operation of a circulator.

4.3 Physical Interpretation and Conclusion

The circulator operates through a non-reciprocal propagation mechanism, meaning that the symmetry between forward and reverse signal transmission is intentionally broken. As a result, the signal is not allowed to propagate equally in both directions, but is instead routed along a preferred cyclic path among the three ports.

The purpose of the two measured configurations is precisely to make this behavior visible experimentally. By changing which pair of ports is connected to the VNA and terminating the remaining port with a matched load, it is possible to isolate the allowed transmission path and compare it with the forbidden one. In this way, the measurements directly show that the circulator routes power preferentially in one direction, while the opposite transmission is suppressed.

This is also the reason why the unused port must always be closed on a matched load: the load absorbs the power reaching that port and prevents it from being reflected back into the device, which would otherwise distort both reflection and transmission measurements.

The measurements confirm the non-reciprocal behavior of the circulator. In both configurations, the comparison between the transmission coefficients clearly shows that one direction is favored, while the opposite one is strongly attenuated. By measuring two different port configurations and properly terminating the third port with a matched $50\,\Omega$ load, it is therefore possible to directly observe the cyclic routing of signals that characterizes the operation of the device. The reflection data additionally indicate that the component remains reasonably well matched over a broad frequency range.

CHAPTER 5

Planar Resonant Structures

5.1 Setup

The device under test (DUT) consists of a planar RF structure enclosed in a metallic housing, containing three different configurations labeled A, B, and C. Each configuration corresponds to a different microstrip layout within the same device and was measured independently.

The DUT was characterized using a Vector Network Analyzer (VNA) in a one-port configuration, measuring the reflection coefficient S_{11}. This choice is dictated by the physical structure of the device, which is accessible from a single port and is designed to operate as a reflective element.

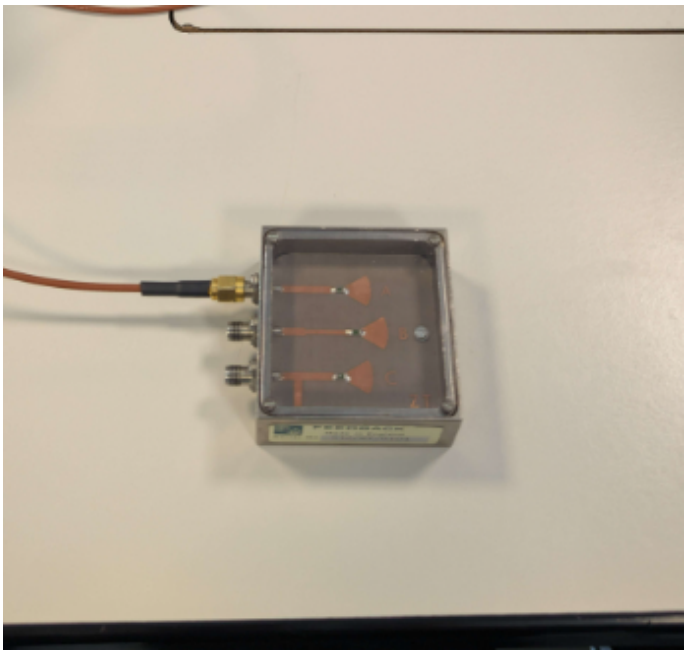

Figure 5.1: Experimental setup showing the DUT connected to the VNA.

5.2 Analysis of the Measurements

The measured $|S_{11}|$ responses for the three configurations are reported in the figures below.

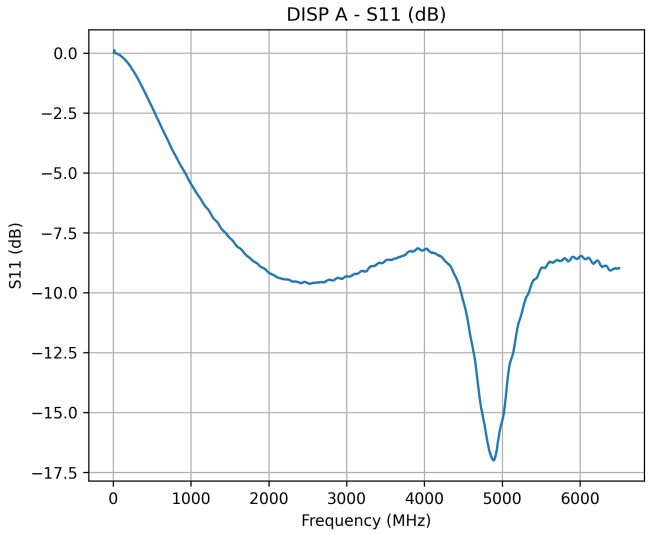

Figure 5.2: S_{11} magnitude for configuration A.

Configuration A exhibits a clear resonance around 5 GHz, visible as a dip in the reflection coefficient reaching approximately −17 dB. The relatively broad shape of the dip indicates a moderate quality factor, consistent with a resonant structure with limited energy confinement and non-negligible losses.

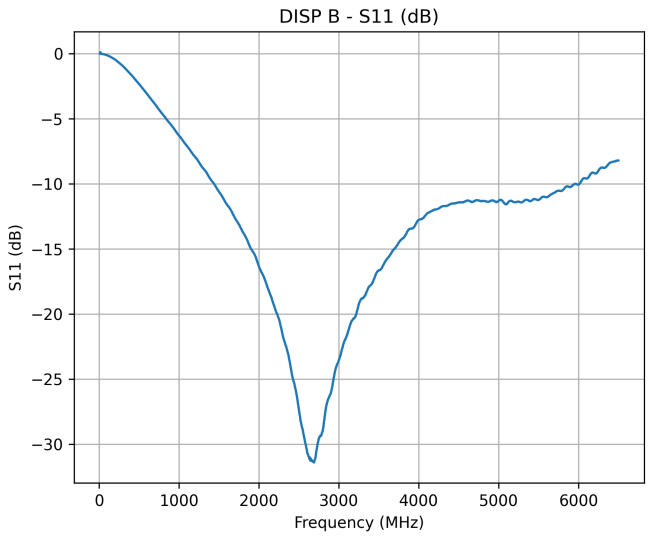

Figure 5.3: S_{11} magnitude for configuration B.

Configuration B shows a single, well-defined resonance around 2.7 GHz, with a much deeper minimum (below −30 dB). This behavior indicates a more efficient impedance matching at resonance and a higher quality factor, meaning that the structure is able to store electromagnetic energy more effectively.

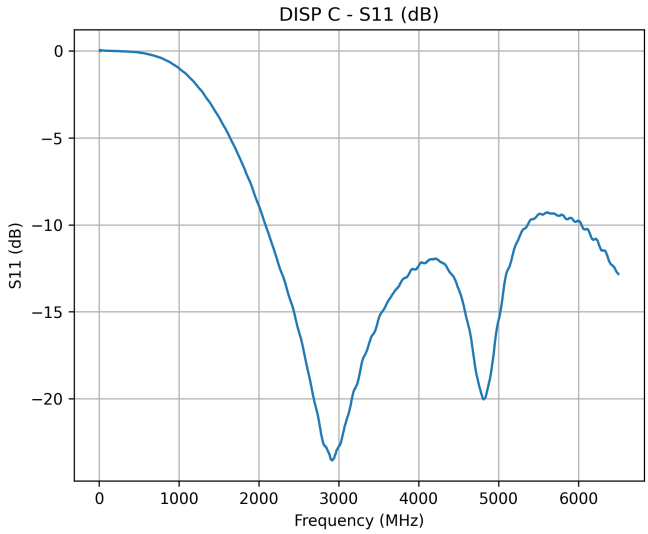

Figure 5.4: S_{11} magnitude for configuration C.

Configuration C exhibits two distinct resonances around 3 GHz and 4.8 GHz. This behavior is a direct consequence of the additional central stub present in the structure, which introduces a second resonant path and coupling between different sections of the device. As a result, multiple resonant modes are supported, each associated with a different effective electrical length and field distribution within the structure.

5.3 Physical Interpretation and Conclusion

The observed responses are characteristic of reflective resonant structures implemented in planar technology, such as microstrip stub resonators.

In all configurations, the dips in S_{11} correspond to frequencies at which the input impedance of the device approaches the characteristic impedance of the measurement system ($50\,\Omega$), resulting in reduced reflection. At these frequencies, the device stores electromagnetic energy, behaving as a resonator.

The variation in resonance frequency between configurations A, B, and C can be attributed to differences in the effective electrical length of the structures:

- Configuration B resonates at the lowest frequency, indicating the largest effective length
- Configuration A resonates at higher frequency, corresponding to a shorter structure
- Configuration C supports multiple resonances, which is most consistently explained by the presence of an additional central stub that introduces extra resonant paths and coupling effects, leading to the excitation of multiple modes

Additionally, the depth and width of the resonant dips provide information about losses and coupling. A deeper and sharper dip, as observed in configuration B, indicates stronger energy confinement and a higher quality factor, while broader resonances, as in configuration A, suggest increased losses or weaker confinement.

Overall, the DUT behaves as a set of planar reflective resonators, with geometry-dependent resonance frequencies and quality factors. The one-port S_{11} measurement is sufficient to identify the resonant behavior and distinguish between different configurations, including single-mode and multi-mode responses.

CHAPTER 6

Band Reject

6.1 Setup

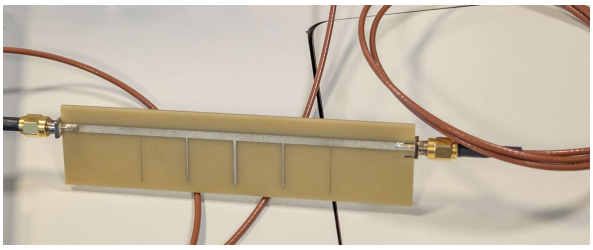

Figure 6.1: Band reject device under test

6.2 Analysis of the measurements

For this device, the plot of the magnitude of the S-matrix covers all the investigated frequency range. This is because the behaviour of the DUT has to be analyzed over a wide frequency band. As a consequence, in Fig. (9) the frequency is reported in GHz.

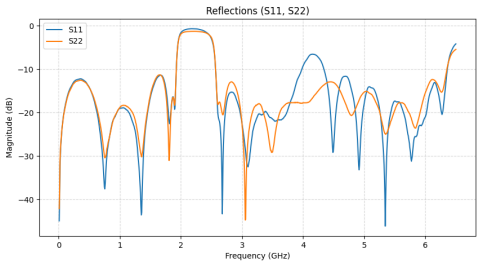

Figure 6.2: Reflection coefficients S11 and S22

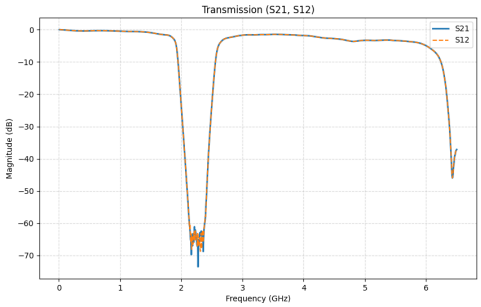

Figure 6.3: Transmission coefficients S21 and S12

The first thing to observe from the plot is that the DUT is passive and (approximately) symmetric, so the behavior is independent of which port is used for transmission.

Then, by observing the transmission plot of S21/S12, we can notice that both at high frequency and at low frequency there is good transmissivity, with the magnitude close to 0 dB, except at a specific frequency where the transmissivity strongly decreases.

Parametro	**Freq (GHz)**	**Magn (dB)**	**BW (GHz)**
S12	2.3140	-68.69	0.8048
S21	2.2750	-73.50	0.8048

Table 6.1: Parametri di trasmissione S12 e S21

With the data processing, we can find the value of the frequency at which the strongest attenuation occurs for both S21 and S12 parameters, reported in Tab. (2). The very low value of about -75 dB guarantees that the signal is almost completely suppressed at that frequency. This is also confirmed by looking at the reflection coefficients S11 and S22, where a peak occurs at the same frequency. The -3 dB bandwidth was estimated to be about 0.8 GHz, corresponding to the frequency region that is not transmitted by the DUT.

6.3 Summary

At the end of the analysis, we can confirm that the DUT is a band-reject filter. To establish the specific frequency we want to suppress from the transmission, the size of the five the vertical lines linked to the main path is fundamental. In fact, to effectively reject a signal, the length of each vertical line, called "stub", must be approximately one-quarter of the wavelength ($\lambda/4$) at the target rejection frequency. Based on our measurements, the resonance occurs at 2.275 GHz. The calculation for the required stub length is as follows:

$$\lambda_0 = \frac{c}{f} = \frac{3 \times 10^8 \text{ m/s}}{2.275 \times 10^9 \text{ Hz}} \approx 131.87 \text{ mm}$$

$$\lambda_g = \frac{\lambda_0}{\sqrt{\epsilon_{\text{eff}}}} \approx \frac{131.87 \text{ mm}}{\sqrt{3.3}} \approx 72.59 \text{ mm}$$

$$L_{\text{stub}} \approx \frac{\lambda_g}{4} \approx \frac{72.59 \text{ mm}}{4} \approx 18.15 \text{ mm}$$

The resulting guided wavelength is in the order of millimeters. When divided by four, the calculated length $L_{stub} \approx 18.15$ mm is perfectly consistent with the physical dimensions of the stubs observed in the device (Fig. 6.1). This confirms that the device was specifically designed to reflect frequencies in the 2.3 GHz range. f